Experiment 4: Design and Comparative Analysis of Butterworth, Chebyshev and Elliptic IIR Filters Using MATLAB.

Program:

```
clc;
clear;
close all;

% EXPERIMENT 4
% Design and Comparative Analysis of Butterworth,
% Chebyshev Type-I and Elliptic IIR Filters

%% STEP 1: INPUT PARAMETERS

Fs  = 1000;     % Sampling frequency in Hz
Fp  = 100;      % Passband edge frequency in Hz
Fst = 150;      % Stopband edge frequency in Hz
Rp  = 1;        % Passband ripple in dB
Rs  = 40;       % Stopband attenuation in dB

% Normalize frequencies with respect to Nyquist frequency

Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

fprintf('========== INPUT PARAMETERS ==========\n');
fprintf('Sampling frequency = %g Hz\n', Fs);
fprintf('Passband edge = %g Hz\n', Fp);
fprintf('Stopband edge = %g Hz\n', Fst);
fprintf('Passband ripple = %g dB\n', Rp);
fprintf('Stopband attenuation = %g dB\n', Rs);
fprintf('Normalized passband frequency = %.3f\n', Wp);
fprintf('Normalized stopband frequency = %.3f\n', Ws);

%% STEP 2: BUTTERWORTH FILTER DESIGN

[NB, WnB] = buttord(Wp, Ws, Rp, Rs);

[bB, aB] = butter(NB, WnB, 'low');

fprintf('\n========== BUTTERWORTH FILTER ==========\n');
fprintf('Filter order = %d\n', NB);
fprintf('Normalized cutoff frequency = %.6f\n', WnB);

disp('Numerator coefficients bB:');
disp(bB);

disp('Denominator coefficients aB:');
disp(aB);

%% STEP 3: CHEBYSHEV TYPE-I FILTER DESIGN

[NC, WnC] = cheb1ord(Wp, Ws, Rp, Rs);

[bC, aC] = cheby1(NC, Rp, WnC, 'low');
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fprintf('\n========== CHEBYSHEV TYPE-I FILTER ==========\n');
fprintf('Filter order = %d\n', NC);
fprintf('Normalized cutoff frequency = %.6f\n', WnC);

disp('Numerator coefficients bC:');
disp(bC);

disp('Denominator coefficients aC:');
disp(aC);

%% STEP 4: ELLIPTIC FILTER DESIGN

[NE, WnE] = ellipord(Wp, Ws, Rp, Rs);

[bE, aE] = ellip(NE, Rp, Rs, WnE, 'low');

fprintf('\n========== ELLIPTIC FILTER ==========\n');
fprintf('Filter order = %d\n', NE);
fprintf('Normalized cutoff frequency = %.6f\n', WnE);

disp('Numerator coefficients bE:');
disp(bE);

disp('Denominator coefficients aE:');
disp(aE);

%% STEP 5: CALCULATE FREQUENCY RESPONSES

[HB, f] = freqz(bB, aB, 4096, Fs);
[HC, ~] = freqz(bC, aC, 4096, Fs);
[HE, ~] = freqz(bE, aE, 4096, Fs);

% Convert magnitude to decibels

MB = 20*log10(abs(HB) + eps);
MC = 20*log10(abs(HC) + eps);
ME = 20*log10(abs(HE) + eps);

% Convert phase to degrees

PB = unwrap(angle(HB))*180/pi;
PC = unwrap(angle(HC))*180/pi;
PE = unwrap(angle(HE))*180/pi;

%% STEP 6: MEASURE RESPONSE AT PASSBAND AND STOPBAND EDGES

% Interpolate the response at Fp and Fst

magB_Fp  = interp1(f, MB, Fp);
magB_Fst = interp1(f, MB, Fst);

magC_Fp  = interp1(f, MC, Fp);
magC_Fst = interp1(f, MC, Fst);

magE_Fp  = interp1(f, ME, Fp);
magE_Fst = interp1(f, ME, Fst);

fprintf('\n========== EDGE FREQUENCY RESPONSE ==========\n');
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```
fprintf('\nButterworth:\n');
fprintf('Magnitude at Fp = %.4f dB\n', magB_Fp);
fprintf('Magnitude at Fst = %.4f dB\n', magB_Fst);

fprintf('\nChebyshev Type-I:\n');
fprintf('Magnitude at Fp = %.4f dB\n', magC_Fp);
fprintf('Magnitude at Fst = %.4f dB\n', magC_Fst);

fprintf('\nElliptic:\n');
fprintf('Magnitude at Fp = %.4f dB\n', magE_Fp);
fprintf('Magnitude at Fst = %.4f dB\n', magE_Fst);

%% STEP 7: DISPLAY ALL GRAPHS IN ONE WINDOW

figure('Name', 'Experiment 4 - IIR Filter Comparison', ...
       'NumberTitle', 'off', ...
       'Color', 'w');

% GRAPH 1: Compare all magnitude responses

subplot(2,2,1);

plot(f, MB, 'b', 'LineWidth', 1.5);
hold on;
plot(f, MC, 'r--', 'LineWidth', 1.5);
plot(f, ME, 'g-.', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');

legend('Butterworth', 'Chebyshev Type-I', ...
       'Elliptic', 'Location', 'best');

xline(Fp, 'k:');
xline(Fst, 'k-.');

xlim([0 Fs/2]);
ylim([-100 5]);

% GRAPH 2: Compare all phase responses

subplot(2,2,2);

plot(f, PB, 'b', 'LineWidth', 1.5);
hold on;
plot(f, PC, 'r--', 'LineWidth', 1.5);
plot(f, PE, 'g-.', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response Comparison');

legend('Butterworth', 'Chebyshev Type-I', ...
       'Elliptic', 'Location', 'best');

xlim([0 Fs/2]);
```

```
% GRAPH 3: Passband comparison

subplot(2,2,3);

plot(f, MB, 'b', 'LineWidth', 1.5);
hold on;
plot(f, MC, 'r--', 'LineWidth', 1.5);
plot(f, ME, 'g-.', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Passband Comparison');

xlim([0 Fp]);
ylim([-3 1]);

legend('Butterworth', 'Chebyshev Type-I', ...
       'Elliptic', 'Location', 'best');

% GRAPH 4: Stopband comparison

subplot(2,2,4);

plot(f, MB, 'b', 'LineWidth', 1.5);
hold on;
plot(f, MC, 'r--', 'LineWidth', 1.5);
plot(f, ME, 'g-.', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Stopband Comparison');

xlim([Fst Fs/2]);
ylim([-100 5]);

legend('Butterworth', 'Chebyshev Type-I', ...
       'Elliptic', 'Location', 'best');

sgtitle('Experiment 4: IIR Filter Design and Comparison');

%% STEP 8: STABILITY CHECK

polesB = roots(aB);
polesC = roots(aC);
polesE = roots(aE);

fprintf('\n========== STABILITY CHECK ==========\n');

disp('Butterworth poles:');
disp(polesB);

disp('Chebyshev Type-I poles:');
disp(polesC);

disp('Elliptic poles:');
disp(polesE);
```

```
fprintf('Butterworth stable: %d\n', all(abs(polesB) < 1));
fprintf('Chebyshev Type-I stable: %d\n', all(abs(polesC) < 1));
fprintf('Elliptic stable: %d\n', all(abs(polesE) < 1));

%% STEP 9: FINAL COMPARISON TABLE

Filter = {'Butterworth'; 'Chebyshev Type-I'; 'Elliptic'};

Order = [NB; NC; NE];

PassbandMagnitude_dB = [magB_Fp; magC_Fp; magE_Fp];

StopbandMagnitude_dB = [magB_Fst; magC_Fst; magE_Fst];

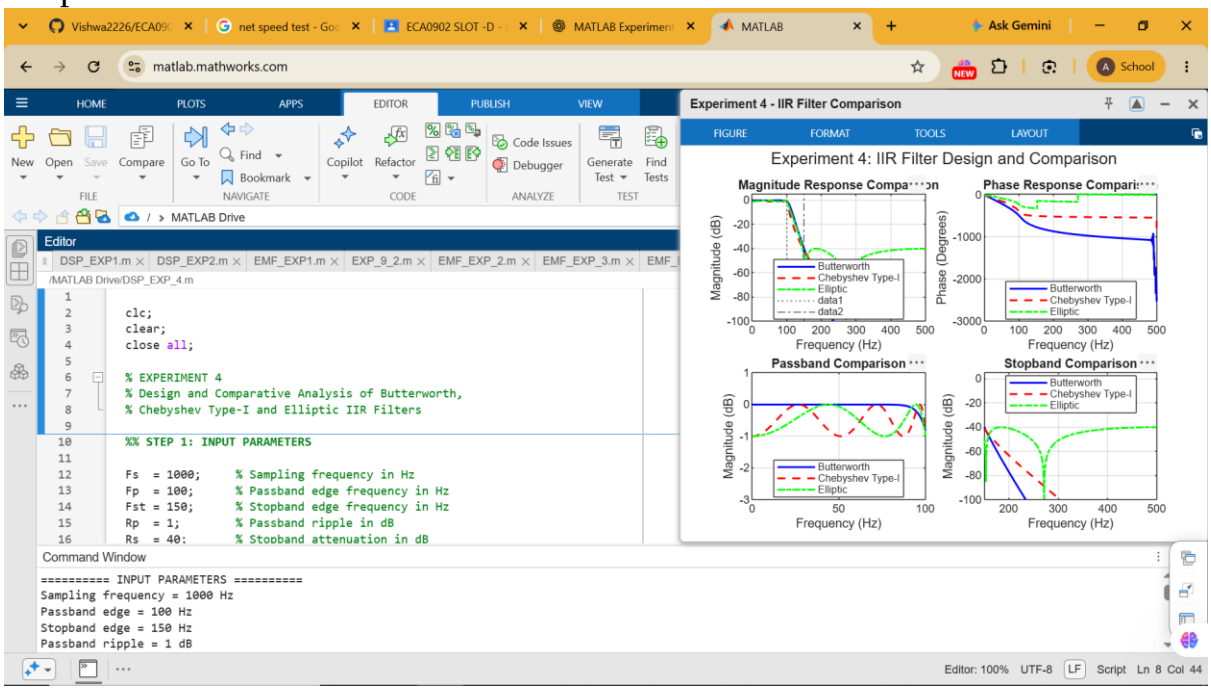
ComparisonTable = table(Filter, Order, ...
    PassbandMagnitude_dB, StopbandMagnitude_dB);

fprintf('\n========== FINAL COMPARISON TABLE ==========\n');
disp(ComparisonTable);

fprintf('Transition width = %.2f Hz\n', Fst-Fp);

disp('Experiment 4 completed successfully.');
```

Output:



Command Window:

========== INPUT PARAMETERS ==========
Sampling frequency = 1000 Hz
Passband edge = 100 Hz
Stopband edge = 150 Hz

Passband ripple = 1 dB
Stopband attenuation = 40 dB
Normalized passband frequency = 0.200
Normalized stopband frequency = 0.300

========== BUTTERWORTH FILTER ==========
Filter order = 12
Normalized cutoff frequency = 0.212709
Numerator coefficients bB:
1.0e-03 *

0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027 0.0002

Denominator coefficients aB:
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899 0.0053

========== CHEBYSHEV TYPE-I FILTER ==========
Filter order = 6
Normalized cutoff frequency = 0.200000
Numerator coefficients bC:
1.0e-03 *

0.0464 0.2782 0.6956 0.9274 0.6956 0.2782 0.0464

Denominator coefficients aC:
1.0000 -4.8694 10.3807 -12.3367 8.5969 -3.3268 0.5586

========== ELLIPTIC FILTER ==========
Filter order = 4
Normalized cutoff frequency = 0.200000
Numerator coefficients bE:
0.0197 -0.0171 0.0333 -0.0171 0.0197

Denominator coefficients aE:
1.0000 -3.0330 3.8118 -2.2911 0.5554

========== EDGE FREQUENCY RESPONSE ==========

Butterworth:
Magnitude at Fp = -0.8080 dB
Magnitude at Fst = -40.0000 dB

Chebyshev Type-I:
Magnitude at Fp = -1.0003 dB
Magnitude at Fst = -41.3237 dB

Elliptic:
Magnitude at Fp = -1.0001 dB
Magnitude at Fst = -48.7165 dB

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========== STABILITY CHECK ==========
Butterworth poles:
   0.7262 + 0.5683i
   0.7262 - 0.5683i
   0.6345 + 0.4627i
   0.6345 - 0.4627i
   0.5699 + 0.3569i
   0.5699 - 0.3569i
   0.5262 + 0.2529i
   0.5262 - 0.2529i
   0.4992 + 0.1508i
   0.4992 - 0.1508i
   0.4862 + 0.0501i
   0.4862 - 0.0501i

Chebyshev Type-I poles:
   0.7820 + 0.5638i
   0.7820 - 0.5638i
   0.8049 + 0.4042i
   0.8049 - 0.4042i
   0.8478 + 0.1486i
   0.8478 - 0.1486i

Elliptic poles:
   0.7621 + 0.5501i
   0.7621 - 0.5501i
   0.7544 + 0.2439i
   0.7544 - 0.2439i

Butterworth stable: 1
Chebyshev Type-I stable: 1
Elliptic stable: 1

========== FINAL COMPARISON TABLE ==========
           Filter           Order    PassbandMagnitude_dB    StopbandMagnitude_dB
    ____________________    _____    ____________________    ____________________

    {'Butterworth'     }     12            -0.80796                 -40
    {'Chebyshev Type-I'}      6             -1.0003               -41.324
    {'Elliptic'        }      4             -1.0001               -48.717

Transition width = 50.00 Hz
Experiment 4 completed successfully.
>>
```